

Open Questions:

1)

a) SQuAD:

It measures reading comprehension by requiring models to answer questions based on a given passage.

It evaluates intrinsic language understanding because the model must understand context, relationships between sentences and semantic meaning in order to locate the correct answer.

b) BoolQ:

BoolQ is a yes/no question answering dataset built from naturally occurring questions and Wikipedia passages.

It measures intrinsic understanding because the model must perform reasoning and comprehension over the text rather than relying only on keyword matching.

c) RACE:

RACE is a multiple-choice QA dataset based on English reading comprehension exams for students.

It evaluates intrinsic language understanding since solving the questions often requires inference, contextual reasoning, and understanding implicit information in the passage.

Practical Part:

- 1) Yes, the configuration that had the highest validation accuracy also produced the best prediction on the test set.
- 2) The worst performing configuration struggled more with sentence pairs that required deeper semantic understanding, especially paraphrases written using wording or sentence structure.
It appeared appeared to rely more on surface-level word similarity causing mistakes on examples where sentences contained similar words but different meaning.

Where the configuration with the best performance handled these cases more successfully because of the more stable training and improved generalization.